

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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1. Product Requirement Document (PRD)

Project: BizMind – AI Business Advisor

Focus: Economic Empowerment & Decision Intelligence

Version 1.0

1.1 Problem Statement

Many young entrepreneurs in Malaysia have the drive to succeed but lack the "data literacy" to grow. They might see that they are making money, but they don't know why some days are better than others. Without expensive consultants or complex accounting software, they rely on "gut feeling," which leads to inconsistent profits and missed growth.

1.2 Proposed Solution

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BizMind is a Decision Intelligence terminal. It takes simple sales data and business descriptions and turns them into a "Virtual COO" report. By combining hard math (backend calculations) with soft reasoning (Z AI GLM), it tells the owner exactly what to do next.

1.3 Target Users

Small Business Owners: Specifically those running stalls, cafes, or home-based businesses.

Online Sellers: People selling on platforms like Shopee or TikTok who need to understand their weekly trends.

1.4 Core Functionalities

The Analysis Engine: A form where users input their business context and sales.

Decision Framework: The system outputs results in a 4-pillar format: Insight (The Fact), Recommendation (The Action), Explanation (The Reason), and Impact (The Goal).

Smart Partner Matching: A system that suggests collaborators (e.g., matching a food seller with a delivery service) based on service descriptions.

2. System Analysis Documentation (SAD)

Objective: To explain the technical architecture and data flow of the BizMind ecosystem.

2.1 System Architecture

We chose a Client-Server Architecture to keep the frontend light and the backend powerful.

Frontend (UI): Built with HTML/CSS and Vanilla JavaScript. It's designed to be clean and "terminal-like" to keep the focus on the data.

Backend (Logic Layer): Powered by FastAPI (Python). This is the "brain" of the app. It doesn't just pass text to the AI; it performs the mathematical calculations first.

Intelligence Layer (Z AI GLM): We use the GLM API as our reasoning engine. It receives the processed data from our backend to generate human-readable advice.

2.2 Data Flow (The Pipeline)

Our system follows a strict Logic-First pipeline:

Ingestion: User inputs sales data (e.g., Weekday RM200, Weekend RM500).

Computation: The Python backend calculates the Variance (150% increase) and identifies the Peak Period.

Context Injection: We "wrap" these numbers into a prompt. Instead of the AI guessing, we tell it: "The user has a 150% increase on weekends. Why might this happen for a cafe?"

Inference: The GLM generates the strategy.

Matching: The system scans businesses. Json using keyword matching to find potential partners.

2.3 Technology Stack

Language: Python (Backend), JavaScript (Frontend), HTML, CSS.

Framework: Fast API (Fast and asynchronous).

AI Model: Z AI GLM API.

Data Storage: JSON (For the business partner database).

3. Decision Intelligence Report (The "Why")

Why is BizMind different from Other AI platforms?

If you ask a general AI "How is my business doing?", it doesn't have your data. If you paste your data into a general AI, it might make mistakes in the math.

BizMind is different because:

Reliable Math: Our Python backend handles the calculations. The AI never has to "do math," which reduces hallucinations.

Structured Output: We force the AI to follow the Insight → Action workflow. This ensures the user isn't just reading a "chat" but a professional business report.

Contextual Awareness: By matching the user with real businesses in our JSON database, we move from "Advice" to "Actionable Partnership."

4. Project Limitations & Future Work

Data Source: Currently uses manual input. Future versions could integrate with POS (Point of Sale) systems.

Matching Algorithm: Currently uses keyword matching. We plan to upgrade this to "Vector Embeddings" for more accurate partnership suggestions.

Financial Scope: Focuses on sales velocity; future updates will include profit margin and cost-of-goods analysis.